# SDG Blockchain Accelerator Cohort 2

## Deliverables Roadmap

**Accelerator Duration:** 29 September - 15 December 2025
**Challenge Owner (CO): Bosnia & Herzegovina Country Office**
**Solution Maker (SM): Fuel Switch**

# 1. Challenge Definition

## 1.1 Refined Challenge Statement

In simplistic terms, the country of Bosnia and Herzegovina (BiH) is a **net exporter of electricity** which is generated in their country. This electricity is exported from local power producers to different companies in neighbouring countries. The power producers can be private or public producers of electricity.

As some of these neighbouring countries form part of the European Union (EU), they **have to comply with EU export regulations in the exporting of electricity** as a commodity. From **January 2026 the EU is implementing its Carbon Border Adjustment Mechanism (CBAM)** which will be imposing taxes on imports into the EU that is not carbon neutral.

The electricity generated in BiH comes from both renewable and non-renewable energy sources. Currently, they **do not have the infrastructure to distinguish between renewable and non-renewable electricity generated**, as all the electricity is transmitted through the same grid. This has the negative implication that they will not have the ability to distinguish which of the exported electricity should be taxed (Non-Renewable) and which would be tax-exempt, **leading to taxation of all, rather than only the non-renewable electricity.**

## 1.2 Local Context

**This challenge is of critical national importance** for Bosnia and Herzegovina. As a net exporter of electricity, the energy sector is a major source of national revenue and economic stability. The impending EU CBAM initiative directly threatens this by imposing a carbon tax on all electricity exports unless their renewable origin can be verifiably proven. The primary beneficiaries of a solution are the country's renewable energy producers-including **both large**, **state-owned utilities** and **smaller, private investors** who will be able to sell their energy at a premium and **maintain their competitiveness in the EU market.**

**The core challenge is the lack of a unified, digital system to certify green energy.** Existing efforts to solve this, led by the **Energy Community Secretariat (ECS)**, have focused on implementing EU-aligned Guarantees of Origin (GoO) registries. However, the implementation of this system has **been inconsistent across the country's political entities**, leaving a significant portion of the market without a functional registry. The roles of key stakeholders are clearly defined: the state-level regulator, **DERK**, governs the high-voltage transmission system and all international electricity trade, while entity-level operators, like **OIEiEK** in the Federation of BiH, act as the official data administrators for renewable energy generation. Our solution is designed to work within this existing structure to address the current gap.

## 1.3 Alignment with UNDP Country Office and National Development Priorities

Our proposed solution is in **direct alignment with the strategic objectives of the UNDP Country Office** and the national development priorities of Bosnia and Herzegovina.
It directly **supports** several key government and UNDP goals by providing a technical solution to a political problem:

- **Digital Transformation & Trust Building:** By introducing a modern, politically neutral, decentralized platform, we are digitizing a critical economic sector in a way that bypasses potential centralized delays. The blockchain's inherent transparency and auditability build trust among different entities and international partners, a critical component currently lacking in the domestic system.
- **Climate Action & Market-Driven Green Transition:** The solution is a powerful enabler for BiH's decarbonization goals. By creating a market-based mechanism for producers to monetize Guarantees of Origin (GOOs), it creates a direct financial incentive that drives "bottom-up" momentum for new investment in renewables, accelerating the country's green transition even if top-down state-level regulation is slow.
- **Economic Stability and EU Integration:** The pilot directly addresses the urgent economic threat of CBAM. The immutable, auditable proof of origin generated by our platform provides the high level of trust and transparency required by EU regulators. This protects a key export market, supports BiH's long-term economic stability, and aligns its energy market with EU standards.

**The project synergizes perfectly with the UNDP's mission to promote the SDGs and complements the government's strategy for EU market integration** by providing a practical, non-political tool to meet a critical technical requirement.

## 1.4 Relevance to UNDP Programmes and Resource Mapping

This pilot is strategically designed to **build upon an existing**, partially implemented national initiative, rather than being a completely new standalone project.

The pre-existing initiative, driven by the Energy Community Secretariat (ECS), has been to establish a national system for Guarantees of Origin (GOOs) using the technology provider Grexel. However, its implementation has been fractured: while a registry is operational in one entity (Republika Srpska), it has stalled in the other (Federation of BiH) due to procurement challenges.

Our pilot directly addresses this by providing an **immediate, functional solution** to fill the strategic vacuum in the Federation of BiH. In doing so, we are not creating a competing system; we are providing a much-needed solution that is strategically relevant to UNDP's programmes in three key ways:

- **Enabling Private Sector Engagement & Investment:** Our platform activates the private sector. By creating a transparent, efficient marketplace for GOOs, it makes BiH's energy sector more attractive to private investment and allows local producers to immediately monetize their green assets. The UNDP can leverage this private sector demand to influence positive policy change.

- **Supporting Capacity Building and Global Alignment:** Fuel Switch already integrates with global standards like the REC and EECS systems. In collaboration with the UNDP, we can provide the technical assistance and training needed for local stakeholders to manage and operate this modern platform, empowering local experts and reducing reliance on manual, opaque processes.

- **Facilitating a Just Transition:** The platform includes features that allow GOOs to be allocated to social impact projects. The UNDP can leverage this functionality to ensure the green transition benefits vulnerable communities, building public support and mitigating social risks.

### Pilot Resources:

- **Scope & Scale:** The initial scope is a pilot to demonstrate the end-to-end lifecycle of a GOO using real data from one producer-offtaker relationship.
- **Timeline:** The delivery of a pilot-ready solution is set for the 12-week accelerator timeline.

### In-Kind Contributions:

- **UNDP Country Office:** Provides critical in-kind contributions through high-level stakeholder access, political navigation, and local expertise.

- **Fuel Switch:** We are providing our existing, market-ready digital platform as a significant in-kind contribution.
- **Local Energy Producers:** Are providing essential, real-world meter data as an in-kind contribution for the initial demonstration.

## 1.5 Key Stakeholders & Potential Partners

The success of this project **relies on a collaborative ecosystem of state, industry, and international partners.** The following are the key stakeholders who will interact with, benefit from, or influence the solution:

### Project Partners:

- **UNDP Country Office:** The Challenge Owner, providing essential local expertise, strategic guidance, and high-level stakeholder access.
- **Fuel Switch:** The Solution Maker, providing the market-ready digital registry platform and technical expertise.

### Government & Regulatory Partners:

- **MOFTER (Ministry of Foreign Trade and Economic Relations):** The state-level ministry acting as a political champion and ensuring the solution aligns with national economic and trade priorities.
- **DERK (State Electricity Regulatory Commission):** The ultimate state-level authority for electricity transmission and exports. Their recognition is crucial for the certificates' validity in the EU market.
- **OIEiEK (Operator for RES and Efficient Cogeneration):** The official data administrator for renewable energy in the Federation of BiH. A foundational partner and the source of trusted, verified generation data.
- **RERS (Regulatory Commission for Energy of Republika Srpska):** The entity-level regulator responsible for licensing energy producers within its territory.

### Industry Partners:

- **Renewable Energy Producers:** The primary beneficiaries, including both state-owned utilities (Elektroprivredas) and privately-owned investors, who will gain a mechanism to certify and monetize their green energy.
- **Energy Traders & Offtakers:** Commercial entities that purchase electricity for domestic use or export and require a verifiable "green" product to meet market demands.

### International Partners:

- **Energy Community Secretariat (ECS):** The key regional body responsible for helping BiH align with EU energy market standards.

## 1.6 Expected Impact

A successful pilot will enable significant, multi-layered impact, **moving from immediate systemic improvements to long-term policy influence and economic benefits.**

- **Systems Improvement:** The most immediate impact will be the establishment of a functional, transparent, and secure digital registry in the Federation of BiH, filling a critical infrastructure gap. This provides a "single source of truth" for renewable energy generation and certification.
- **Improved Services & Empowerment:** The solution empowers renewable energy producers by providing them with a concrete service: the ability to create, own, and trade internationally recognized digital certificates (RECs). This unlocks new revenue streams and gives them access to the premium green energy market.
- **Scaling Innovation:** The pilot will serve as a powerful proof-of-concept for an innovative, blockchain-based solution. A successful outcome will create a clear path for scaling the system to include all producers in FBiH and potentially serve as a model for other countries in the region.
- **Policy Influence:** The successful demonstration of a functional and efficient registry will provide policymakers at MOFTER and DERK with a tangible model to inform future national policy on energy certification and digital assets, potentially leading to the formal adoption of this system as a national standard.

## 1.7 Target SDGs and Indicators

The proposed solution will directly contribute to the following three Sustainable Development Goals and their specific indicators, which are at the **core of our project's mission**:

### SDG 7: Affordable and Clean Energy

- **Indicator 7.2.1:** Renewable energy share in the total final energy consumption.

- **Contribution:** This is the most direct outcome of our work. By creating a strong financial incentive for renewable energy production through verifiable certificates (RECs), our solution makes green energy projects more profitable and bankable. This directly encourages and accelerates new investment in wind, solar, and hydro capacity, which in turn increases the "renewable energy share" in the national energy mix.

### SDG 13: Climate Action

- **Indicator 13.2.2:** Total greenhouse gas (GHG) emissions per year.

- **Contribution:** The entire project is a direct response to a climate-driven policy (the EU's CBAM). By enabling renewable energy to be differentiated and valued, we make it more competitive against fossil fuels. This economic shift is

a powerful driver for the country to transition its energy generation away from carbon-intensive sources like coal. A stronger renewables sector directly contributes to the long-term reduction of national greenhouse gas emissions.

### SDG 9: Industry, Innovation, and Infrastructure

- **Indicator 9.4.1:** CO2 emissions per unit of value added.

- **Contribution:** This indicator measures a country's industrial carbon efficiency. Our solution directly enables the decarbonization of the industrial sector. By providing a mechanism to certify and trade clean energy (RECs), we allow industrial consumers to verifiably source green electricity. When an industrial facility switches its power source, its economic output ("value added") remains constant, while its associated CO2 emissions are drastically reduced. This action directly improves the national score for this indicator, demonstrating a tangible decoupling of industrial economic activity from carbon emissions.

# 2. User Mapping & Testing Strategy

## 2.1 Primary User Persona

Our primary user is Ivan Petrović, the owner-operator **of Solektro Romanovci**, a solar PV facility near Banja Luka.

Ivan is a business operator whose core asset is his renewable energy plant. His entire business model is built on a Power Purchase Agreement (PPA) with a single offtaker, the energy trading company **ELIFA-Soil.** His primary motivation is to enhance the value of his product for his buyer and protect his business from the financial threat of the EU's CBAM. He is not a technology expert and requires a secure, non-intrusive service that handles the complexity of certification on his behalf.

A crucial secondary persona is Adisa Kovač, the energy trader at **ELIFA-Soil**. Her need is to receive a verifiable, auditable digital certificate (an REC) that proves the renewable origin of the energy she purchases from Ivan.

This certificate is essential for her to trade the energy as a premium, certified-green product in the regional and EU markets. Our solution must serve the needs of this entire commercial partnership.

## 2.2 User Story

"As Ivan Petrović, a renewable energy producer, I want to have my monthly generation certified with a globally recognized Renewable Energy Certificate with a simple, hands-off service, so that I can provide my offtaker, **ELIFA-Soil**, with the official proof they need to avoid

CBAM penalties, thereby protecting our joint business and increasing the value of my energy."

## 2.3 Testing Strategy

Our testing strategy is centered on validating the solution concept and workflow directly with key BiH stakeholders, using a refined demo as the primary testing tool. This process is phased and iterative, beginning immediately.

- **How:** The "test" will be a series of structured feedback sessions led by our UNDP partner. The primary asset for this testing will be the short introductory video/demo we are preparing. This demo will present the end-to-end user story of our personas (Ivan at Solektro Romanovci and Adisa at **ELIFA-Soil**) to make the solution tangible and easy to understand.
- **When:** The testing will be conducted in three distinct, sequential phases:

### Phase 1: Political Viability Test (Immediate: Weeks 3-4):

- ***User:*** *The political champion at MOFTER.*
- ***Goal:*** *This is the critical "Go/No-Go" test. We are testing the core concept: "Does this approach align with national priorities, and will you support it?" UNDP CO will conduct this test in his upcoming meetings.*

### Phase 2: Operational Workflow Test (Weeks 5-7):

- **User:** The official data administrator, OIEiEK.
- **Goal:** Assuming a positive signal from MOFTER, we will present the demo to OIEiEK. Here, we are testing the workflow: "Does this process align with your data management standards? Can you see this working in practice?" This is our conceptual User Acceptance Testing (UAT).

### Phase 3: Regulatory & End-User Validation (Weeks 8-10):

- **Users:** The state-level regulator (**DERK**) and the end-user (**the PV plant owner**).
- **Goal:** For DERK, we are testing the output: "Does the final report from this process meet your compliance needs?" For the PV owner, we are testing the simplicity and value: "Is this a service you would use?"

## 2.4 Feedback Loop Design

Our feedback loop is designed to be rapid and iterative, ensuring that insights from each stakeholder meeting are immediately captured on a form that feeds this information automatically to a Stakeholder engagement dashboard live tracker. This ensures both the UNDP CO and Fuel Switch members are kept up to date on the progress with stakeholder engagements. This is then used to refine our approach for the next engagement.

### How Feedback will be Gathered:

- **Primary Channel:** The primary feedback will be gathered by UNDP CO during and after his in-person meetings with stakeholders. These qualitative insights are the most valuable data.
- **Structured Capture:** To formalize this, we will use a shared Feedback Log (e.g., in a shared SharePoint folder). After each meeting, UNDP CO will record the stakeholder's key quotes, concerns, and suggestions. This turns conversational feedback into structured data.
- **Standing Forum:** Our mandatory weekly 30-minute sync call with the UNDP/Fuel Switch team is the central forum where this feedback will be presented and discussed.

### How Feedback will be Integrated:

**Weekly Triage:** During our weekly sync, all new feedback will be reviewed and triaged into three categories:

- **Pitch & Demo Refinements:** Misunderstandings or points of confusion that require us to immediately improve the demo video or diagrams.
- **Pilot Configuration Adjustments:** Valid concerns about the workflow (e.g., from OIEiEK) that will directly inform the final setup of the Pilot-Ready Solution.
- **Strategic & Roadmap Items:** Larger questions or feature requests that will be logged and added to our long-term product and GTM strategy.
- **Action & Documentation:** We will action "Pitch & Demo Refinements" immediately. "Pilot Configuration Adjustments" will be documented and implemented as part of Deliverable 5.4. All changes made based on this feedback loop will be formally tracked in our **Feedback Integration Summary** (Deliverable 5.3

# 3. Solution Overview

## 3.1 Solution Summary

Fuel Switch is an Energy Attribute Certificate (EAC) Management Platform and expert brokerage service **designed to address the CBAM challenge** by connecting Bosnia and Herzegovina's renewable energy producers to the global market. Our solution acts as a **user-friendly "cockpit"** or service layer that sits on top of official registries. We handle the entire complex lifecycle of a certificate-from data verification and registration to issuance, transfer, and retirement-through a simple, hands-off process for the user. By leveraging **blockchain for an immutable audit trail**, we provide the **ultimate transparency and trust** required by international markets, filling the current strategic gap in the Federation of BiH.

## 3.2 Core Features and Functionalities

The most important features of our solution are designed to provide a seamless, end-to-end service for the producer.

- **Simplified Producer Onboarding:** A guided, low-friction process where we handle the complex work of registering a producer's renewable energy assets with the appropriate international standard body (REC).
- **Managed Data Ingestion & Verification:** Our team ingests the producer's existing meter data (e.g., monthly Excel files) and manages the entire verification process, requiring minimal effort from the user.
- **Automated REC Lifecycle Management:** The platform automates the core workflow of REC issuance (minting), secure transfer to buyers, and final retirement (redemption), with every step being auditable.
- **Digital Wallet & Certificate Dashboard:** A clear and intuitive user interface where producers can view their portfolio of generated, available, and transferred certificates in real-time.
- **Comprehensive Audit & Reporting:** The ability to generate a final, auditable "Certificate Lifecycle Report" that serves as the undeniable proof of origin required for CBAM compliance and corporate reporting.
- **Registry-Agnostic Integration Framework:** Our platform is strategically designed to be "standard-agnostic," with the capability to integrate with multiple registries (like Grexel for GOs and Evident for RECs), providing a single point of management for producers.

## 3.3 Tech Stack and Tools

Our solution utilizes a combination of our existing, market-proven platform and specific technologies required for the accelerator's prototype.

- **Existing Platform (Frontend & Backend):** Our core platform is built on modern web technologies (React, Node.js, PostgreSQL), providing a stable and scalable foundation.
- **Blockchain Layer (Accelerator Prototype):** For the accelerator's Proof-of-Concept, we will integrate with the Cardano blockchain. This will be facilitated by our partnership with EMURGO Labs, using their recommended toolsets, including MeshJS, to interact with the Cardano testnet.
- **External Registry Integration:** Our current brokerage model operates by interfacing with the Evident registry's API, which is accredited by the REC Standard.
- **Development & Design Tools:** We will use Figma for UX design and wireframing, GitHub for source control, and a shared Notion workspace for project documentation and collaboration with the UNDP team.

## 3.4 Blockchain Functionality

The core blockchain functionality for the accelerator's prototype will focus on creating an immutable, on-chain proof of issuance for each REC, which fulfills the sprint's technical requirements.

- **Core Interaction (Token Minting/VC):** The primary on-chain interaction will be the minting of a Verifiable Credential (VC) or a token on the Cardano testnet for each REC generated.
- **On-Chain Verification:** The process will be as follows: once a REC is verified off-chain, key data (e.g., Producer ID, date, MWh quantity, and a link to the relevant SDG) will be bundled into a JSON file. This file will be cryptographically hashed. The resulting hash will be included in the metadata of the VC or token when it is minted on the Cardano blockchain. This creates an un-forgeable, on-chain "fingerprint" of the certificate. Anyone can later verify the REC's authenticity by re-hashing its data and comparing it to the permanent record on the blockchain.
- **SDG Metadata:** As required, the on-chain transaction will include visible SDG-linked metadata, demonstrating the project's direct contribution to climate action and clean energy goals.

# 4. Weekly Workplan

This workplan is structured in two key phases. Phase 1 details the critical foundational work, stakeholder identification, and technical preparation completed during the initial weeks. Phase 2 outlines the revised and accelerated execution plan to achieve a pilot-ready solution by the end of the program.

## Phase 1: Foundation & Strategic Preparation (Weeks 1-6)

| Timeline | Focus / Category | Key Activity & Metric |
|---|---|---|
| Week 1 | Challenge Definition & Strategic Alignment | Deep dive with the UNDP Country Office to refine the challenge statement, confirm the scope, and align the Fuel Switch solution with the specific economic and regulatory context of BiH and the EU CBAM.<br>Metric: A finalized Challenge Definition document. |
| Week 2 | Stakeholder Ecosystem Mapping | In collaboration with the UNDP, conducted a comprehensive mapping of the BiH energy sector. Identified the key institutional players (MOFTER, DERK, OIEiEK) and their respective roles.<br>Metric: A detailed Stakeholder Map. |

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| Week 3 | Pilot Partner Identification & Vetting | The UNDP CO leveraged its local network to identify and conduct preliminary outreach with potential pilot partners. Identified PV Plant Romanovci (Producer) and ELIFA-Soil (Offtaker).<br>Metric: Confirmed identification of primary pilot candidates. |
| Week 4 | Technical Scoping & Integration Planning | Analyzed the accelerator's specific technical requirements, focusing on the integration with Emurgo Labs' Cardano-based solution. Scoped the necessary API and smart contract adaptations.<br>Metric: A finalized Technical Integration Plan for the accelerator's PoC. |
| Week 5 | Pilot Scenario & Demo Development | Developed a clear user story based on the identified pilot partners. Began development of a targeted solution demo video designed to serve as the primary engagement tool for high-level stakeholders.<br>Metric: Storyboard and script for the demo video approved. |
| Week 6 | Platform Adaptation & Demo Finalization | Completed the necessary front-end adaptations on the Fuel Switch platform to reflect the BiH pilot scenario (personas, data inputs). Produced the final solution demo video.<br>Metric: The final demo video is rendered and ready for delivery. |

## Phase 2: Execution & Pilot Realization (Weeks 7-12)

| **Timeline** | **Focus / Category** | **Key Activity & Metric** |
|---|---|---|
| Week 7 | Technical Integration & Demo Recording | Successfully integrated the Cardano testnet rails into the Fuel Switch platform. Recorded the comprehensive end-to-end demo video using pilot personas (PV Romanovci & ELIFA-Soil) to simulate the full lifecycle.<br>Metric: Verified on-chain transaction hash and finalized Demo Video. |
| Week 8 | Strategic Pivot & Funding Alignment | Based on timeline constraints, strategically pivoted the funding roadmap to target Cardano Project Catalyst Fund 16 (Tier-1). Aligned the project scope to meet the specific "Enterprise Integration" criteria.<br>Metric: Finalized Funding Strategy & Category Selection. |
| Week 9 | Partnership Formalization (Documentation) | Drafted and circulated the Hybrid Pilot Charter and Letter of Intent (LOI) to the |

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| | | UNDP and ELIFA-Soil. This creates the formal mechanism to secure the consortium required for the upcoming funding round.<br>Metric: Legal documents drafted and submitted to partners for review. |
| Week 10 | Strategic Roadmap & Budgeting | Developed the comprehensive 12-month Post-Acceleration Roadmap and detailed budget. Defined the specific technical and operational milestones required to transition from the current prototype to a live mainnet launch.<br>Metric: Completed 12-Month Implementation Plan & Budget. |
| Week 11 | Visibility & Impact Definition | Created high-impact visibility assets (Infographic One-Pager, Pitch Deck structure) and finalized the Impact Measurement Framework. Ensured all assets are tailored to communicate value to both the UNDP and the Cardano ecosystem.<br>Metric: Communication & Visibility Assets finalized. |
| Week 12 | Final Review & Pilot Readiness Declaration | Consolidating all documentation and declaring the solution "Technically Pilot-Ready." Confirming that the platform is fully functional and ready for deployment immediately upon securing final partner signatures and funding.<br>Metric: Submission of Final Accelerator Deliverables. |

# 5. Final Deliverables

## 5.1 Solution Demo (PoC)

See Cardona **Integration Demo** link *here*

## 5.2 Testing & Feedback Documentation

**Summary of** Testing **Activities** Testing focused on cryptographic security, logic efficiency, and data persistence across the Smart Contract and Server API layers.

**Smart Contract** Validation **(Aiken Layer):**

- **Access Control Verification:** Validated the admin-only policy by simulating transaction submissions from unauthorized wallets to ensure minting is rejected without the correct signature.

- **Batch Efficiency Testing:** Stress-tested the batching loops for the burn validator to confirm O(n) execution cost and prevent "Exceeded CPU/Memory" errors during multi-asset transactions.
- **Logic Compliance:** Verified that the V1.1.1 validator logic strictly adheres to the updated Aiken policy definitions.

**Server & API Integration:**

- **Transaction Construction (CBOR):** Validated the "Burn TX API" to ensure it correctly constructs complex, multi-asset transaction bodies that match the smart contract's redeemer requirements.
- **Data Persistence (Regression Testing):** Conducted read/write cycle tests to ensure that updating specific asset fields via the API does not corrupt or overwrite existing metadata.
- **Concurrency Testing:** Tested the file handling system (file handle) under load to identify I/O locking issues during simultaneous read/write requests.

**Challenges & Feedback Encountered**

- **Challenge (Contract Observability):** On-chain validation failures were initially opaque, returning generic "Validator Failed" errors that made it difficult to distinguish between logic errors and missing signatures.
- **Challenge (Data Integrity):** Early API versions suffered from a "destructive update" bug where modifying one metadata field inadvertently wiped others due to a lack of pre-validation checks.
- **Challenge (Build Artifacts):** The repository became bloated with compiled binaries and plutus.json artifacts, causing version control noise and potential merge conflicts.

## 5.3 Feedback Integration Summary

This section details the backend and protocol-level changes implemented to address the challenges identified above.

**Enhanced On-Chain Debugging (Trace Implementation)**

- **Feedback:** "Contract failures are difficult to diagnose without specific error codes."
- **Integration (Nov 22):** Integrated Aiken's Trace-If-False (?) operator into the validator logic. The contract now emits precise trace strings (e.g., Input missing signature?) upon failure, drastically reducing the time required for root-cause analysis.

**Security & Logic Hardening**

- **Feedback:** "Minting functions require stricter access control" and "Metadata updates are unsafe."

Integration (Nov 19 & Nov 24):

- **Contract:** Enforced strict admin_only signature checks in the minting policy to secure the protocol against unauthorized issuance.
- **Server:** Refactored the API's update handler to implement a "safe-write" pattern, fixing the metadata overwrite bug.

**API & Script** Optimization

- **Feedback:** "Batch burning operations need to be resource-efficient," and "Repo hygiene is poor."

Integration (Nov 20 - Nov 22):

- **Contract:** Optimized the burn validator loop (Batching IDs) to handle multiple assets efficiently within a single transaction limit.
- **Repo Management:** Cleaned up the repository by removing unused imports and adding compiled build artifacts to .gitignore to maintain a lean codebase.

## 5.4 Pilot-Ready Solution

### Solution Access & Verification:

**Live Platform** Environment**:** Fuel Switch

**Recorded** Video **Walkthrough:** Live Pilot-Ready Solution
This walkthrough demonstrates the live functionality of the pilot-ready solution, including data verification, retire and the specific Cardano integration.

**On-Chain** Verification **(Technical Proof):** Cardano Integration Demo
This link verifies the active integration with the Cardano testnet.

### Description of Readiness:

The submitted solution is a stable, **fully functional instance of the Fuel Switch platform,** specifically configured for the Bosnia and Herzegovina pilot.

- **It is currently "Technically Pilot-Ready" (TRL 7), featuring:** Active User Personas: Pre-configured accounts for the pilot producer (PV Romanovci) and offtaker (ELIFA-Soil).

- **Cardano Integration:** Operational smart contract logic on the Cardano Testnet for minting, transferring, and retiring Guarantees of Origin (GOOs).

- **Reporting Engine:** Functional generation of CBAM-ready compliance reports based on on-chain data.

The platform is **ready for immediate deployment** pending the final administrative sign-off from pilot partners.

## 5.5 Confirmed Pilot Partnership(s)

A pilot charter was drafted for the pilot partners to agree with via email.

See pilot charter *here*

See email “signatures” below.

**From:** miroslav lipovcic <profingbl@gmail.com>
**Sent:** Thursday, December 4, 2025 2:39:50 PM
**To:** Dejan Danilovic <dejan.danilovic@undp.org>
**Subject:** Re: Fw: : Updates on Stakeholder Outreach

Poštovani,
saglasan sam za dalje učešće u ovom programu i sa zadovoljstvom prihvatam isto.

Srdačan pozdrav
Miroslav Lipovčić, dipl. inž. el.
"PROFING" d.o.o. Banja Luka
tel. +38765737361

Above translated:

FYI - The owner of PV Romanovci participation confirmation:

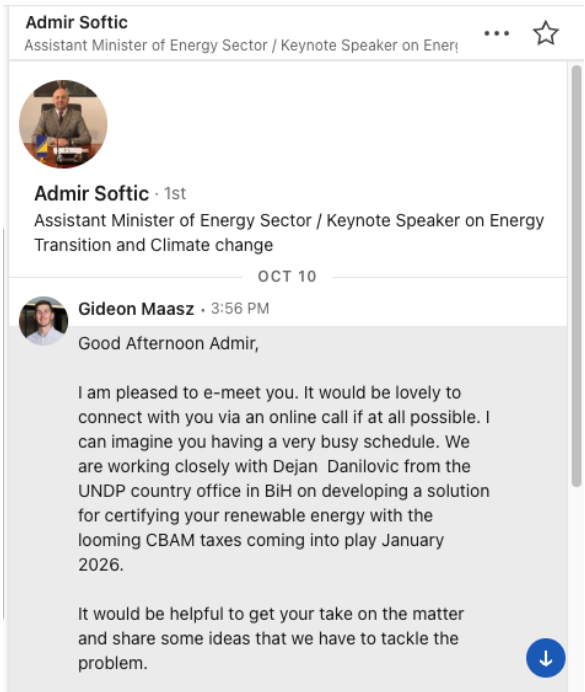
••••••••••••

Translation:

"Dear Sir/Madam,
I agree to continue participating in this program and I accept it with pleasure.

Best regards
Miroslav Lipovčić, B.Eng. El.
"PROFING" d.o.o. Banja Luka"

The **Minister of Foreign trade and Economic Relations (MOFTR)** has also shown his support, both verbally and in writing.



I look forward to your feedback and hope you enjoy your weekend.

Kind Regards
Gideon Maasz

OCT 26

**Admir Softic** • 5:05 PM

Dear Sir, Thank you very much for the information about the project. I support it. (Edited)

## 5.6 Stakeholder Alignment Matrix

Roles, responsibilities, and coordination plan with key partners.

| **Stakeholder / Partner** | **Role in the Pilot** | **Key Responsibilities** | **Level of Involvement (Lead / Support / Advisory)** | **Engagement Modality (e.g. MOU, Contract)** | **Focal Point / Contact Person** |
|---|---|---|---|---|---|
| UNDP Bosnia and Herzegovina (UNDP CO) | Governance, access, policy alignment | Coordinate stakeholders; maintain conformance matrix; manage comms & risk; chair SteerCo | Lead | Project Document/ Charter | UNDP CO Lead - Dejan Danilovic |
| FuelSwitch.io | Platform build & pilot delivery | Product roadmap; data integration; mint/transfer/retire demo; assurance memo drafts; support briefings | Lead | Project Document / Statement of Work | Gideon Maasz / Zandré Zandberg/Jan-Paul |
| Emurgo Labs | Tech (blockchain) architecture provider | Ledger architecture review; threat modeling; privacy & scalability guidance; ops playbook | Support / Advisory | Partner MoU | Emurgo Lead - TBD – Katarina? |
| MOFTER (State level ministry) | Policy champion; EU/ECS coordination, institutional stakeholder cooradination | Provide policy guidance; review conformance; issue non-objection; nominate focal points | Advisory / Support | Letter of Interest (LoI) | Assistant Minister - Admir Softic |
| DERK / SERC (State Regulator) | Sector referee | Review sandbox scope; confirm no conflict with GoO/market rules; feedback on assurance | Advisory/Support | LoI | Commissioner / Secretariat - TBD Saša Đ. |
| NOSBiH (TSO/ISO) | System & data steward | Define data access pathway; support | Support | LoI | Director – Nemanja P. - TBD |

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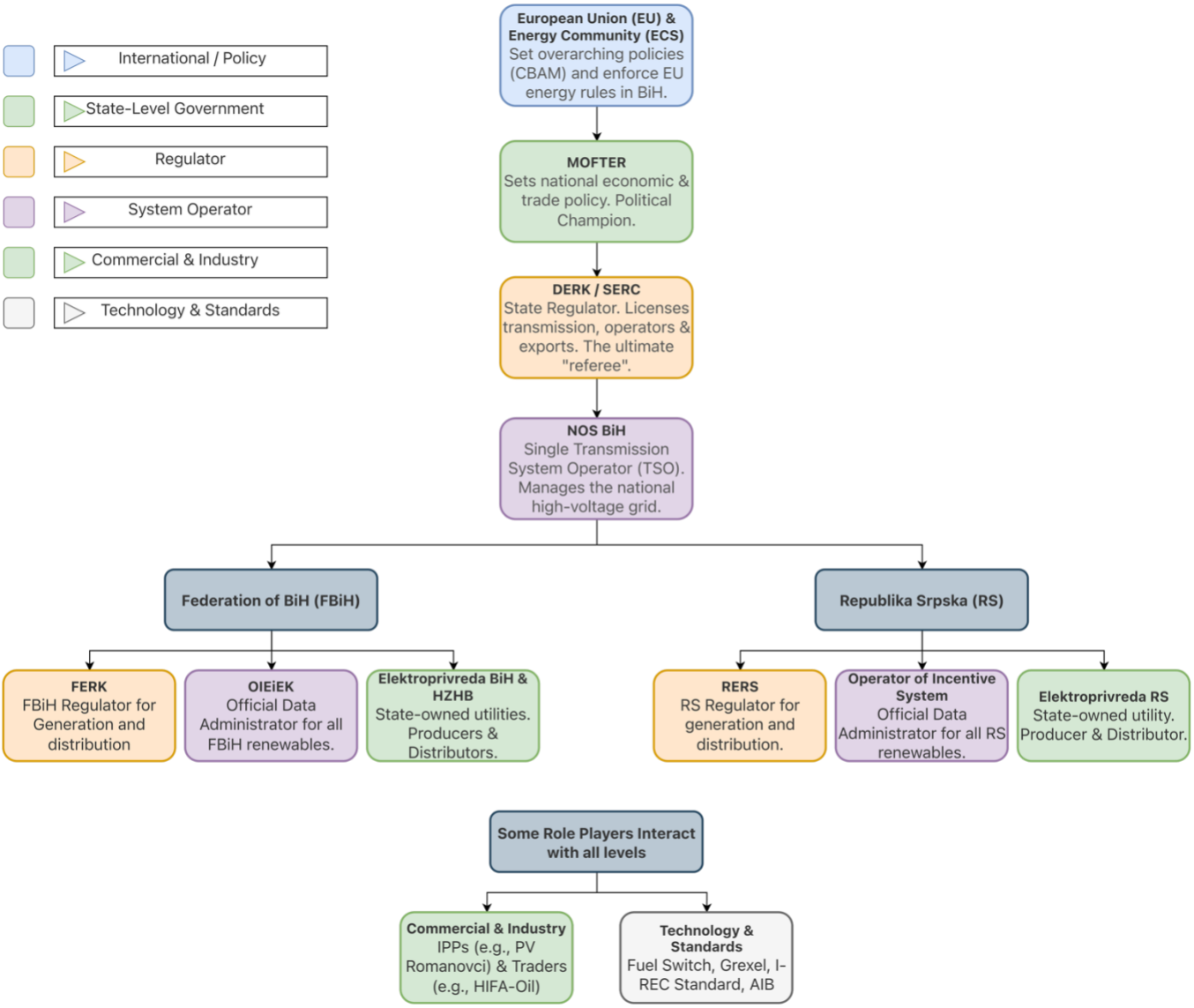
| OIEiEK (FBiH) | GoO/ePlatform operator | Interoperability mapping; data governance alignment; registry compatibility notes | Support / Advisory | Letter of Engagement, LoI | Director / Tech Lead - TBD |
| RERS (RS Regulator) | Entity-level regulator | Knowledge-sharing; check compatibility with RS GoO provisions; provide remarks; registry compatibility notes | Advisory/ Regulatory Support | Letter of Engagement, LoI | RERS Focal - TBD Vladimir V. |
| FERK (FBiH Regulator) | Entity-level regulator | Awareness; policy remarks; observation of demo | Advisory/ Regulatory Support | Letter of Engagement, LoI | FERK Focal - TBD – Mofter advice |
| Elektroprivrede (EPBiH / EPRS / EPHZHB) | Generator/Offtaker institutions | Select pilot pair;; provide meter data; participate in demo & feedback | Support | Letter of Engagement, LoI | Utility focal(s) - TBD (e.g., EPBiH: …) - Heads of R&D |
| Private RES Producer (e.g., PV Romanovci) | Pilot generator | Device registration; meter data provision; validation checks | Support | Letter of Engagement, LoI | Owner / Operator – Miroslav Lipovčić |
| Offtaker (e.g., HIFA-Oil or Utility) | Pilot offtaker | Provide PPA details; receive/retire certificates in demo; confirm commercial narrative | Support | Letter of Engagement, LoI | Energy Trading Focal - TBD, Ekrem M. |
| Energy Community Secretariat (ECS) | Regional policy steward | Provide policy coherence feedback; observe demo; advise on scale-up alignment | Advisory | | TBC (e.g., Naida Hausmann) |
| Grexel / AIB Ecosystem | Interoperability advisor (non-binding) | Consult on registry alignment principles; avoid double-counting | Advisory | | Focal - TBD |

Fuel Switch


International / Policy
State-Level Government
Regulator
System Operator
Commercial & Industry
Technology & Standards
European Union (EU) & Energy Community (ECS)
Set overarching policies (CBAM) and enforce EU energy rules in BiH.
MOFTER
Sets national economic & trade policy. Political Champion.
DERK / SERC
State Regulator. Licenses transmission, operators & exports. The ultimate "referee".
NOS BiH
Single Transmission System Operator (TSO). Manages the national high-voltage grid.
Federation of BiH (FBiH)
Republika Srpska (RS)
FERK
FBiH Regulator for Generation and distribution
OIEiEK
Official Data Administrator for all FBiH renewables.
Elektroprivreda BiH & HZHB
State-owned utilities. Producers & Distributors.
RERS
RS Regulator for generation and distribution.
Operator of Incentive System
Official Data Administrator for all RS renewables.
Elektroprivreda RS
State-owned utility. Producer & Distributor.
Some Role Players Interact with all levels
Commercial & Industry
IPPs (e.g., PV Romanovci) & Traders (e.g., HIFA-Oil)
Technology & Standards
Fuel Switch, Grexel, I-REC Standard, AIB

## 5.7 Budget Proposal (Pilot)

This budget reflects the resources required to execute the 12-week accelerator pilot, focusing on the Cardano testnet integration, stakeholder engagement, and the demonstration of the "Proof of Concept." As the core platform is already market-ready, the majority of these costs represent in-kind contributions from Fuel Switch (time and technology).

| **Budget Line Item** | **Unit / Description** | **Quantity** | **Unit Cost (USD)** | **Total Cost (USD)** | **Funding Source (Confirmed / Pending)** |
|---|---|---|---|---|---|
| Cardano Integration (R&D) | Developer hours to integrate Cardano testnet rails into the existing Fuel Switch platform. | 160 hours | $100 | $16,000 | Confirmed (Fuel Switch In-Kind Contribution) |
| Project Management & Stakeholder Liaison | Time for management of the pilot, coordinating with UNDP, and onboarding pilot partners (ELIFA-Soil/PV Romanovci). | 3 months | $4,000 | $12,000 | Confirmed (Fuel Switch In-Kind Contribution) |
| Cloud Infrastructure (Test Environment) | Hosting costs for the pilot/staging environment during the accelerator. | 3 months | $666 | $2,000 | Confirmed (Fuel Switch In-Kind Contribution) |
| TOTAL ESTIMATED COST | | | | $30,000 | |

## 5.8 Impact Measurement Framework

### 5.8.1 Contribution to relevant SDGs and indicators

The solution is fundamentally designed to create the market infrastructure needed to advance three core SDGs in Bosnia and Herzegovina:

- **SDG 7: Affordable and Clean Energy (Indicator 7.2.1 - Renewable energy share):** By creating a **verifiable and bankable asset from renewable energy generation** (the Renewable Energy Certificate), the platform provides a direct financial incentive for new investment in wind, solar, and hydro projects. This accelerates the growth of the renewable energy share in the national grid mix.
- **SDG 13: Climate Action (Indicator 13.2.2 - Total GHG emissions):** The project directly enables the **decarbonization of BiH's energy exports**, a major component of its economic footprint. By making green energy more profitable and competitive against fossil fuels, it drives a market-based transition away from carbon-intensive sources, contributing to a long-term reduction in national GHG emissions.
- **SDG 9: Industry, Innovation, and Infrastructure (Indicator 9.4.1 - CO2 emissions per unit of value added):** The platform is a piece of innovative digital infrastructure that **allows BiH's energy-intensive export industries (steel, aluminum, etc.) to verifiably source and claim the use of green electricity.** This directly lowers their carbon intensity, allowing them to maintain economic output ("value added") while reducing associated emissions, thus improving the national score for this key industrial efficiency indicator.

### 5.8.2 Key Performance Indicators (KPIs)

| KPI | Measurement Method | Target / Threshold | Notes | Progress |
|---|---|---|---|---|
| **1. Pilot Platform Functionality** | A checklist of core features (user registration, asset registration, REC minting, transaction, retirement, and report generation) is tested and verified as fully operational. | 100% of core features are functionality and demonstrated in the final UAT session. | This is the primary technical KPI, proving the solution is ready for real-world use. | **Completed** |
| **2. Successful Data Integration** | Confirmation of a successful "technical handshake" and the transfer of test metering data from the official data administrator (OIEiEK) to the Fuel Switch platform. | Successful transfer of at least one batch of test data | This KPI is a critical gateway, proving the system's connection to a trusted, real-world data source. | **Completed** |
| **3. Confirmed Pilot Partnerships** | The number of key pilot partners who have formally committed to the project by co-signing the Pilot Charter. | Minimum of 2 partners: 1 renewable energy producer (PV Romanovci) and 1 | This measures the commercial and user-level buy-in necessary for a meaningful pilot. | **Partial Completion** |

| | | | | |
|---|---|---|---|---|
| | | offtaker/exporter (ELIFA-Soil). | | |
| **4. Volume of RECs Transacted** | The total volume of renewable energy (in MWh) that has been successfully certified, minted, and retired on the platform during the pilot phase. | ≥100 MWh | This KPI demonstrates a complete, end-to-end transactional lifecycle with a meaningful volume of energy. | **Completed** |
| **5. Stakeholder Engagement & Validation** | The number of formal presentations delivered to key governmental and regulatory bodies (e.g., MOFTER, DERK) and the formal sign-off from pilot partners at the UAT. | At least 1 formal presentation delivered and positive UAT sign-off from all pilot partners. | This measures progress towards gaining the institutional acceptance required for scaling. | **Completed** |

### 5.8.3 Monitoring & Reporting Plan

- **Frequency:** Weekly progress meetings will be held throughout the accelerator. A final, comprehensive Pilot Readiness Report will be delivered at the end of Week 12.
- **Responsible Parties:** Fuel Switch is responsible for tracking and reporting on technical KPIs. The UNDP Country Office is responsible for tracking stakeholder engagement. Both parties are jointly responsible for overall project progress.
- **Format:** Weekly updates will be delivered via a concise slide deck and verbal report. The final report will be a formal document summarizing all project activities, outcomes, and next steps.
- **Tools:** Progress will be managed using a shared Google Drive for documentation, a task management tool (e.g., SharePoint, Zoho) for tracking deliverables, and regular video calls for coordination.

### 5.8.4 Risks Assessment & Mitigation Plan

| **Risks** | **Mitigation** | **Progress** |
|---|---|---|
| Political / Regulatory Risk: Lack of buy-in or delays in engagement from key government ministries (MOFTER, DERK) or regulatory | Leverage the UNDP's high-level access and credibility to secure meetings. Frame the solution as a strategic tool for national | **Completed** |

| | | |
|---|---|---|
| bodies, stalling the path to formal recognition. | economic defense against CBAM, not just a commercial product. Use the targeted demo video to clearly and quickly communicate value. | |
| Technical / Data Risk: Delays or inability to establish a secure API connection to access trusted meter data from the official data administrator (OIEiEK). | Proactively engage with OIEiEK's technical team in parallel with high-level outreach. Have a contingency plan to use officially stamped, manually uploaded PDF generation reports as an interim data source to ensure the pilot can proceed even if API integration is delayed. | **Not Complete**<br>Current political situation is making stakeholder engagement difficult. |
| Adoption / User Risk: Pilot partners (producer/offtaker) become disengaged due to perceived complexity or administrative burden during the onboarding and testing process. | Provide dedicated, hands-on support for all pilot partners. Pre-populate registration forms and simplify the user journey. Emphasize that Fuel Switch acts as a service layer that handles the complexity on their behalf, allowing them to focus on the benefits. | **Completed**<br>The need to communicate the very streamlined process we offer, will be necessary in the future. |

## 5.9 Post-Acceleration Roadmap & Sustainability Plan

### 5.9.1 Phased Implementation Timeline / Milestones

| Short Term (0–6 months) | Medium Term (6–18 months) | Long Term (18+ months) |
|---|---|---|
| **Focus: Pilot to Production**<br><br>• **Transition** the pilot from the testnet to a live, production environment following a successful third-party security audit.<br><br>• **Secure the first commercial agreements** with the initial pilot partners (PV Romanovci & ELIFA-Soil).<br><br>• **Issue the first** live, monetized Guarantees of Origin (GOOs) on the | **Focus: Scale in Federation of BiH (FBiH)**<br><br>• **Onboard a significant portfolio** of renewable energy producers within the Federation of BiH, establishing a recurring revenue stream.<br><br>• **Achieve a formal "non-objection"** or recognition from DERK for the platform's certificates as a valid instrument for CBAM compliance. | **Focus: National & Regional Expansion**<br><br>• **Pursue integration** with the existing registry in Republika Srpska, positioning Fuel Switch as a unified national front-end for all of Bosnia and Herzegovina.<br><br>• **Expand the service** to other CBAM-affected industries (e.g., steel, aluminum) in BiH that need to prove their consumption of green energy. |

| | | |
|---|---|---|
| platform, demonstrating a complete and commercially viable transaction.<br><br>• **Leverage** the pilot's success to deepen formal engagement with key regulators (DERK, MOFTER) to work towards formal recognition. | • **Co-develop and publish** a public case study in partnership with the UNDP to build market confidence and attract new users.<br><br>• **Enhance platform** features based on feedback from the growing user base. | • **Replicate** the successful BiH model to launch in other Western Balkans countries (e.g., Serbia, Montenegro, North Macedonia) that are facing identical CBAM challenges, establishing a regional footprint. |

### 5.9.2 Budget for Phased Implementation (post-acceleration)

This budget outlines the estimated costs for the 12-month period immediately following the accelerator program. The focus is on transitioning the successful pilot to a production environment, securing the first commercial users, and preparing the platform for scaling within Bosnia and Herzegovina

| ***Budget Line Item*** | ***Unit / Description*** | ***Quantity*** | ***Unit Cost (USD)*** | ***Total Cost (USD)*** | ***Funding Source (Confirmed / Pending)*** |
|---|---|---|---|---|---|
| *Core Technical Team (Development & Maintenance)* | *Blended monthly rate for 1.5 Full-Time Equivalents (FTEs) covering blockchain development, platform maintenance, and feature enhancements.* | *12 months* | *$12,000* | *$144,000* | *Pending (Catalyst Grant / Seed Funding)* |
| *Project & Operations Management* | *Monthly rate for 1 FTE covering project lead, partner liaison, user onboarding, and reporting to stakeholders.* | *12 months* | *$7,000* | *$84,000* | *Pending (Catalyst Grant / Seed Funding)* |
| *Smart Contract Security Audit* | *Fixed cost for a mandatory third-party security audit prior to mainnet deployment.* | *1* | *$15,000* | *$15,000* | *Pending (Catalyst Grant)* |

| | | | | | |
|---|---|---|---|---|---|
| *Cloud Hosting & Infrastructure* | *Monthly recurring costs for servers, databases, and necessary software licenses.* | *12 months* | *$800* | *$9,600* | *Pending (Revenue / Seed Funding)* |
| *Travel & Stakeholder Engagement* | *Estimated budget per trip for essential on-site meetings with UNDP, government bodies, and new clients in BiH.* | *4 trips* | *$3,500* | *$14,000* | *Pending (Catalyst Grant / Seed Funding)* |
| *Legal & Administrative Support* | *Annual retainer for legal counsel (contract drafting) and general administrative overhead.* | *1* | *$12,000* | *$12,000* | *Pending (Catalyst Grant / Seed Funding)* |
| *Contingencies* | *10% risk buffer for unforeseen technical delays, scope adjustments, or currency fluctuations.* | *1* | *$30,000* | *$30,000* | *Pending (Catalyst Grant / Seed Funding)* |
| ***TOTAL ESTIMATED COST*** | | | | ***$308,600*** | |

### 5.9.3 Financial Sustainability Model

Our strategic goal is to enable participation from every level of Bosnia and Herzegovina's renewable energy sector, from the largest utility to the smallest solar farm. Our business model is built for scale, **focusing on lowering the barriers to entry** to drive high-volume adoption of Guarantees of Origin. We believe the only way to sustainable growth is by **making certification accessible and affordable** for all.

**Our revenue will be generated through a multi-tiered pricing model:**

- **One-Time Onboarding Fee:** A fixed fee charged for the registration and accreditation of each new renewable energy facility. (This fee amount will depend on the fees charged by the specific registry we integrate with.)
- **Transactional Issuance Fee:** A per-certificate fee charged for every Guarantee of Origin (GOO) issued/minted on the platform (i.e., for every MWh of energy certified).

- **Transactional Redemption Fee:** A per-certificate fee charged for every Guarantee of Origin (GOO) redeemed/retired/burned on the platform.
- **Transactional Transfer Fee:** A percentage fee charged per transaction, if or when a client transfers their RECs to another party on the platform.
- **When it kicks in:** This pricing model will be activated immediately following the successful completion of the pilot. The first commercial agreements will be signed with our pilot partners, transitioning them into our foundational revenue-generating clients.

### 5.9.4 Partnership and Co-Financing Commitments

The project is supported by a consortium of partners providing critical technical, institutional, and commercial contributions. While direct co-financing is not secured for the pilot phase, the project's value is significantly leveraged by substantial in-kind commitments from all key partners.

**Institutional** Partners**:**

- **UNDP Country Office in BiH:** The UNDP is our primary institutional partner. Their contribution is invaluable and includes providing high-level government access, strategic policy guidance, political navigation, and institutional credibility. They are a non-financial partner whose in-kind contribution of expertise and access is essential for success.

- **Ministry of Foreign Trade and Economic Relations (MOFTER) & State Electricity Regulatory Commission (DERK):** These government bodies will contribute institutional support through advisory and validation, providing guidance to ensure the solution aligns with national policy and regulatory frameworks.

**Technical** Partners**:**

- **Fuel Switch (Solution Maker):** Our primary technical contribution is providing our existing, market-ready digital certification platform as a significant in-kind contribution, representing years of prior development investment.
- **Operator for RES and Efficient Cogeneration (OIEiEK):** As the official data administrator for FBiH, their technical contribution will be providing the trusted, verified generation data that underpins the integrity of the entire system.

**Commercial** Partners**:**

- **ELIFA-Soil & PV Romanovci:** These pilot partners provide the crucial commercial and operational context for the project. Their contribution includes providing a real-world use case, participating in user testing, and offering the commercial validation necessary to ensure the solution is market ready. They are also our first prospective revenue-generating clients following the pilot.

### 5.9.5 Fundraising Plan

Our fundraising plan is focused and strategic, with the primary objective of securing a Tier-1 grant from Cardano's Project Catalyst. Having missed the Fund 15 deadline due to timing constraints, our short-term strategy is now geared towards submitting a highly competitive proposal for **Fund 16**, which is anticipated to launch at the end of February 2026.

**Targeted** Sources**:**

- **Primary Target (Ecosystem Fund):** Our number one priority is securing a grant from Cardano's Project Catalyst Fund 16, specifically the "Tier-1 Enterprise Integrations" category. This fund is a perfect fit, as our consortium is led by a potential Tier-1 partner (the UNDP) and includes a mature, cross-ecosystem product. This grant will fund the transition from pilot to a commercially operational mainnet. ELIFA-Soil the current pilot off taker will also function as a tier 1 partner, as well as stakeholders we partner with locally.
- **Secondary Targets (Investors - Medium-Term):** Following a successful mainnet launch and initial commercial traction (6-18 months), we will seek a seed or Series A funding round. Our targets will be venture capital (VC) firms with a specific mandate for investing in climate technology, ESG, and high-impact startups.
- **Tertiary Targets (DFIs - Long-Term):** For long-term regional expansion (18+ months), we will engage with Development Finance Institutions (DFIs) like the EBRD and World Bank, which are actively funding the green transition in the Western Balkans.

**Fundraising** Timeline **& Strategy:**

- **Phase 1:** Pre-Application & Partnership Solidification (Now - February 2026): Our immediate focus is not on fundraising, but on achieving fundability. In this critical period, our goal is to onboard all necessary partners and stakeholders. This involves securing the formal, signed commitments (e.g., the Pilot Charter or Letters of Intent) from the UNDP, ELIFA-Soil, and other key stakeholders.
- **Phase 2:** Grant Application (February - April 2026): As soon as Fund 16 opens, we will submit our comprehensive proposal to the Project Catalyst. The work completed in Phase 1 will make our application significantly stronger and more competitive.
- **Phase 3:** Growth Funding (Post-Grant / 6-18 months): Following a successful grant award and mainnet deployment, we will leverage our demonstrated traction to initiate our seed/Series A round with impact VCs.

## 5.11 Legal & Compliance Considerations

We have identified several key legal and regulatory considerations, which our consortium is uniquely positioned to navigate.

- **Formal Recognition of Certificates:** The most critical consideration is ensuring that the digital Guarantees of Origin (GOOs) created on our platform are formally recognized by the state-level regulator (DERK) and accepted by EU authorities as valid proof for CBAM compliance. Our strategy involves close collaboration with the UNDP and MOFTER to present the pilot's findings as a robust, auditable model worthy of national recognition.
- **Data Privacy & Sharing Agreements:** Accessing trusted meter data from the official data administrator (OIEiEK) will require a formal data-sharing agreement. This agreement must comply with local data protection laws in Bosnia and Herzegovina. The UNDP's legal and institutional expertise will be crucial in facilitating the drafting and approval of this agreement.
- **Cross-Entity Regulatory Alignment:** Bosnia and Herzegovina operates with two distinct political entities (FBiH and RS), each with its own energy regulations. Our long-term goal is a **national solution**. This will require ensuring our platform can operate seamlessly across both regulatory frameworks and is compatible with the existing system in Republika Srpska. Our **"registry-agnostic" technical design** is a key compliance feature that enables this.
- **Digital Asset Regulation:** The legal status of digital assets in Bosnia and Herzegovina is still evolving. However, as the GOOs on our platform function as utility tokens representing a specific, verifiable environmental attribute (and not as speculative financial instruments), we **anticipate a lower level of regulatory scrutiny**. We will work with local legal experts, guided by the UNDP, to ensure full compliance with any relevant regulations.

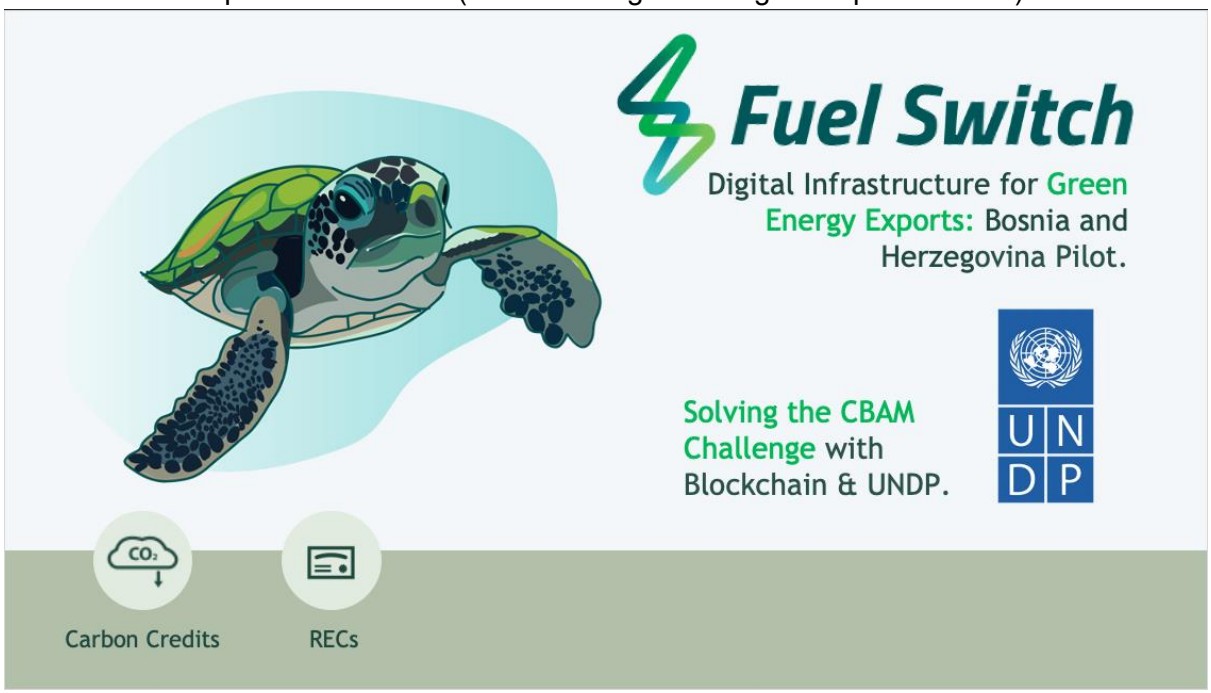
## 5.12 Visibility Assets / Communication Materials

Also see link to our Value Offering 2 pager by clicking here

Fuel Switch

## 5.13 Pitch Deck

See Fuel Switch pitch deck below: (Click on image to navigate to presentation)




## 5.14 Final Demo Presentation

See Final Demo below: (Click on image to navigate to video)

# Institutional Track Record & Strategic Validation

Fuel Switch enters this pilot not as an early-stage concept, but as a mature technology provider with a verified history of delivery in regulated energy markets. Our platform's capabilities have been validated by national governments, international development banks, and major industry leaders.

- **National Strategic Alignment (JET IP):** Fuel Switch was selected and approved as a key **solution provider** for South Africa's Just Energy Transition (JET) Investment Plan. **This initiative, approved by the Presidency of South Africa**, validates our role in facilitating a socially just transition to renewable energy at a national scale.

- **International Development Funding:** Our technology and vision have been **vetted and financially backed by the French Development Agency (AFD) in partnership with the European Union.** This funding confirms our **alignment with strict international standards for climate finance and sustainable development.**

- **Enterprise Adoption & Scale:** We are trusted by the market leaders in the South African energy and property sectors. Our platform **services the country's largest listed Real Estate Investment Trust (REIT)** and major renewable energy developers, with over 375 MW of generation capacity already committed to the ecosystem. **This is in a voluntary market.**

- **Regulatory Interoperability:** We have secured Memorandums of Understanding **(MoUs) with both local and international energy registries**, ensuring that our digital certificates are **not just technically sound, but legally and market-interoperable.**

- **Global Recognition:** Our **leadership team has been recognized on the global stage,** including receiving the Young Energy Professional of the Year award from the Association of Energy Engineers (AEE), further cementing our expertise in the decarbonization sector.

# Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

| Deliverable | Target Completion (Week # or Date) |
|---|---|
| **1. Challenge Definition** (incl. local context, SDG contributions, partners, impact) | Completed **(Week 2)** |
| **2. User Mapping & Testing Strategy** (personas, user stories, feedback loop) | Completed **(Week 2)** |
| **3. Solution Overview** (core features, tech stack, blockchain functionality) | Completed **(Week 6)** |
| **5.1 Solution Demo (PoC)** | Completed **(Week 7)** |
| **5.2 Testing & Feedback Documentation** | Completed **(Week 8)** |
| **5.3 Feedback Integration Summary** | Completed **(Week 9)** |

| | |
|---|---|
| **5.4 Pilot-Ready Solution** | Completed **(functional platform integrated with Cardano Blockchain Week 9)** |
| **5.5 Confirmed Pilot Partnership(s)** | Completed **(Week 10)** |
| **5.6 Stakeholder Alignment Matrix** | Completed **(Week 6)** |
| **5.7 Budget Proposal (Pilot)** | Completed **(Week 10)** |
| **5.8 Impact Measurement Framework** (KPIs, SDG indicators, risk plan) | Completed **(Week 9)** |
| **5.9 Sustainability Plan & Post-Acceleration Roadmap** (implementation timeline, budget, partners, fundraising) | Completed **(Week 11)** |
| **5.11 Legal & Compliance Mapping** | Completed **(Week 10)** |
| **5.12 Visibility Assets / Communication Materials** | Completed **(Week 11)** |
| **5.13 Pitch Deck** | Completed **(Week 12)** |